

SSIE 553 - Homework 4

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Question 1

Given the LP:

$$\text{Max } z = -x_1 + x_2$$

Subject to:

$$2x_1 + x_2 + s_1 = 4 \quad \dots (1)$$

$$x_1 + x_2 + s_2 = 2 \quad \dots (2)$$

$$x_1, x_2, s_1, s_2 \geq 0$$

where s_1 and s_2 are slack variables. The basic variables are x_2 and s_1 .

Part (i): Determine x_B , x_N , B , N , c_B , c_N , and b

We have four variables: x_1, x_2, s_1, s_2 . The basic variables (BV) are x_2 and s_1 . The non-basic variables (NBV) are x_1 and s_2 .

Basic and Non-Basic Variable Vectors:

$$x_B = [x_2, s_1]^T$$

$$x_N = [x_1, s_2]^T$$

Basis Matrix B (columns of basic variables from the constraint matrix):

From the constraints in standard form:

$$\text{Constraint 1: } 2x_1 + x_2 + s_1 = 4$$

$$\text{Constraint 2: } x_1 + x_2 + s_2 = 2$$

The column of x_2 is $[1, 1]^T$ and the column of s_1 is $[1, 0]^T$.

$$B = \begin{bmatrix} | & x_2 & s_1 & | & & 1 & 1 & | \\ | & & & | & = & | & & | \\ | & x_2 & s_1 & | & & 1 & 0 & | \end{bmatrix}$$

Non-Basis Matrix N (columns of non-basic variables):

The column of x_1 is $[2, 1]^T$ and the column of s_2 is $[0, 1]^T$.

$$N = \begin{bmatrix} | & x_1 & s_2 & | & & 2 & 0 & | \\ | & & & | & = & | & & | \\ | & x_1 & s_2 & | & & 1 & 1 & | \end{bmatrix}$$

Objective Function Coefficients:

From $z = -x_1 + x_2 + 0s_1 + 0s_2$:

$$c_B = [c_{x_2}, c_{s_1}]^T = [1, 0]^T$$

$$c_N = [c_{x_1}, c_{s_2}]^T = [-1, 0]^T$$

Right-Hand Side Vector b :

$$b = [4, 2]^T$$

Part (ii): Construct the Simplex Tableau

To construct the simplex tableau for the given basis, we need B^{-1} , then compute $B^{-1}N$, $B^{-1}b$, and the reduced costs.

Step 1: Compute B^{-1}

$$B = \begin{bmatrix} | & 1 & 1 & | \\ | & & & | \\ | & 1 & 0 & | \end{bmatrix}$$

$$\det(B) = (1)(0) - (1)(1) = -1$$

$$B^{-1} = \begin{bmatrix} 0 & -1 \\ -1 & 1 \end{bmatrix} / (-1) = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 2: Compute $B^{-1} * b$ (basic variable values)

$$B^{-1} * b = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} * \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

So $x_2 = 2$ and $s_1 = 2$.

Step 3: Compute $B^{-1} * N$

$$B^{-1} * N = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} * \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 4: Compute the reduced costs for non-basic variables

Reduced cost = $cN^T - cB^T * B^{-1} * N$

$$cB^T * B^{-1} * N = [1, 0] * \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = [1, 1]$$

$$\text{Reduced costs} = cN^T - cB^T * B^{-1} * N = [-1, 0] - [1, 1] = [-2, -1]$$

Step 5: Compute z-value

$$z = cB^T * B^{-1} * b = [1, 0] * [2, 2]^T = 2$$

The Simplex Tableau:

Setting $x_N = 0$ (non-basic variables at zero):

BV	x1	x2	s1	s2	RHS
x2	1	1	0	1	2
s1	1	0	1	-1	2
z	-2	0	0	-1	2

Note: The z-row shows the reduced costs (coefficients in the z-row for each variable). A negative reduced cost means increasing that variable would decrease z.

Part (iii): Comment on Feasibility and Optimality

Feasibility:

A basis is feasible if $B^{-1} * b \geq 0$ (all basic variable values are non-negative).

$$B^{-1} * b = [2, 2]^T$$

Both values are non-negative ($x_2 = 2 \geq 0$ and $s_1 = 2 \geq 0$), so the basis IS FEASIBLE.

Optimality:

For a maximization problem, a feasible basis is optimal if all reduced costs of non-basic variables are ≤ 0 .

$$\text{Reduced cost of } x_1 = -2 \leq 0 \quad (\text{OK})$$

$$\text{Reduced cost of } s_2 = -1 \leq 0 \quad (\text{OK})$$

All reduced costs are non-positive, so the basis IS OPTIMAL.

Conclusion:

The given basis $\{x_2, s_1\}$ is both feasible and optimal. The optimal solution is $x_1 = 0$, $x_2 = 2$, $s_1 = 2$, $s_2 = 0$, with optimal objective value $z = 2$.

Question 2

Given the LP:

$$\text{Max } z = 3x_1 + 7x_2 + 5x_3$$

Subject to:

$$x_1 + x_2 + x_3 \leq 50 \quad (\text{Sugar})$$

$$2x_1 + 3x_2 + x_3 \leq 100 \quad (\text{Chocolate})$$

$$x_i \geq 0$$

Optimal basic variables: $\{x_3, x_2\}$. Non-basic variables: $\{x_1, s_1, s_2\}$.

Given:

$$B^{-1} = \begin{vmatrix} & 3/2 & -1/2 & | \\ & & & | \\ & -1/2 & 1/2 & | \end{vmatrix}$$

First, let us identify B and compute key quantities.

The constraint matrix in standard form (with slacks s_1, s_2):

$$x_1 + x_2 + x_3 + s_1 = 50$$

$$2x_1 + 3x_2 + x_3 + s_2 = 100$$

Basic variables are $\{x_3, x_2\}$, so:

$$B = \begin{vmatrix} & x_3 & x_2 & | & & 1 & 1 & | \\ & & & | & = & & & | \\ & x_3 & x_2 & | & & 1 & 3 & | \end{vmatrix}$$

Compute $B^{-1} * b$ (optimal basic variable values):

$$B^{-1} * b = \begin{vmatrix} & 3/2 & -1/2 & | & 50 & | \\ & & & | & & | \\ & -1/2 & 1/2 & | & 100 & | \end{vmatrix}$$

$$x_3 = (3/2)(50) + (-1/2)(100) = 75 - 50 = 25$$

$$x_2 = (-1/2)(50) + (1/2)(100) = -25 + 50 = 25$$

So the optimal solution is $x_1 = 0, x_2 = 25, x_3 = 25$.

$$z = 3(0) + 7(25) + 5(25) = 0 + 175 + 125 = 300$$

Compute $cB^T * B^{-1}$ (shadow prices / dual variables):

$$cB = [c_{x_3}, c_{x_2}] = [5, 7]$$

$$cB^T * B^{-1} = [5, 7] * \begin{vmatrix} & 3/2 & -1/2 & | \\ & & & | \\ & -1/2 & 1/2 & | \end{vmatrix}$$

$$= [5(3/2) + 7(-1/2), 5(-1/2) + 7(1/2)]$$

$$= [15/2 - 7/2, -5/2 + 7/2]$$

$$= [4, 1]$$

So $y_1 = 4$ (shadow price for sugar) and $y_2 = 1$ (shadow price for chocolate).

Part (i): Range of c_1 (Type 1 candy bar profit)

For the current basis to remain optimal, the reduced cost of x_1 (a non-basic variable) must remain ≤ 0 (for maximization).

Reduced cost of x_1 :

$$r_1 = c_1 - cB^T * B^{-1} * a_1$$

where a_1 is the column of x_1 in the constraint matrix: $a_1 = [1, 2]^T$.

$$cB^T * B^{-1} * a_1 = [4, 1] * [1, 2]^T \\ = 4(1) + 1(2) = 6$$

$$r_1 = c_1 - 6$$

For optimality: $r_1 \leq 0$

$$c_1 - 6 \leq 0$$

$$c_1 \leq 6$$

The current basis remains optimal for $c_1 \leq 6$ (i.e., $-\infty < c_1 \leq 6$).

If $c_1 = 7$:

Since $7 > 6$, the current basis is no longer optimal. We need to pivot x_1 into the basis.

Compute $B^{-1} * a_1$:

$$B^{-1} * a_1 = \begin{array}{ccc|ccc} & 3/2 & -1/2 & | & 1 & | & 1/2 & | \\ & & & | & & | & & | \\ & -1/2 & 1/2 & | & 2 & | & 1/2 & | \end{array}$$

The updated column for x_1 is $[1/2, 1/2]^T$. Both entries are positive.

Minimum ratio test:

$$x_3 \text{ row: } 25 / (1/2) = 50$$

$$x_2 \text{ row: } 25 / (1/2) = 50$$

Tie - we can choose either. Let x_3 leave (first row). x_1 enters, x_3 leaves.

Pivot element = $1/2$. Multiply row 1 by 2:

New row 1 (x_1 row): $[1, 1, \dots]$ with $x_1=50$ (after pivoting)

Actually, let us perform this more carefully using the updated tableau.

New BFS after pivoting (x_1 enters for x_3):

New basis: $\{x_1, x_2\}$

$$B_{\text{new}} = \begin{array}{ccc|ccc} & 1 & 1 & | & & | \\ & & & | & & | \\ & 2 & 3 & | & & | \end{array}$$

$$\det(B_{\text{new}}) = 3 - 2 = 1$$

$$B_{\text{new}}^{-1} = \begin{array}{ccc|ccc} & 3 & -1 & | & & | \\ & & & | & & | \\ & -2 & 1 & | & & | \end{array}$$

$$B_{\text{new}}^{-1} * b = \begin{array}{ccc|ccc} & 3 & -1 & | & 50 & | & 50 & | \\ & & & | & & | & & | \\ & -2 & 1 & | & 100 & | & 0 & | \end{array}$$

So $x_1 = 50$, $x_2 = 0$, $x_3 = 0$.

$$z = 7(50) + 7(0) + 5(0) = 350$$

The new optimal solution when $c_1 = 7$ is: $x_1 = 50$, $x_2 = 0$, $x_3 = 0$, $z = 350$.

Part (ii): Range of c_2 (Type 2 candy bar profit)

x_2 is a basic variable. Changing c_2 affects the reduced costs of ALL non-basic variables. We need all reduced costs of non-basic variables to remain ≤ 0 .

With c_2 changed to c_{2_new} , cB becomes $[5, c_{2_new}]$:

$$cB^T * B^{-1} = [5, c_{2_new}] * \begin{array}{ccc|ccc} & 3/2 & -1/2 & | & & | \end{array}$$

$$\begin{vmatrix} -1/2 & 1/2 \end{vmatrix}$$

Reduced cost of x1 (non-basic):

$$\begin{aligned} r_1 &= c_1 - cB^{\wedge T} * B^{\wedge(-1)} * a_1 \\ &= 3 - [(15-c_2)/2, (c_2-5)/2] * [1, 2]^{\wedge T} \\ &= 3 - [(15-c_2)/2 + 2(c_2-5)/2] \\ &= 3 - [(15-c_2+2c_2-10)/2] \\ &= 3 - [(5+c_2)/2] \\ &= (6-5-c_2)/2 = (1-c_2)/2 \end{aligned}$$

For optimality: $r1 \leq 0 \Rightarrow (1-c2)/2 \leq 0 \Rightarrow c2 \geq 1$

Reduced cost of s1 (non-basic):

Column of s1 in constraint matrix: $a_{s1} = [1, 0]^T$.

$$\begin{aligned} r_{s1} &= 0 - [(15-c_2)/2, (c_2-5)/2] * [1, 0]^T \\ &= -(15-c_2)/2 = (c_2-15)/2 \end{aligned}$$

For optimality: $r_{s1} \leq 0 \Rightarrow (c2-15)/2 \leq 0 \Rightarrow c2 \leq 15$

Reduced cost of s2 (non-basic):

Column of s2: $a_{s2} = [0, 1]^T$.

$$\begin{aligned} r_{s2} &= 0 - [(15-c_2)/2, (c_2-5)/2] * [0, 1]^T \\ &= -(c_2-5)/2 = (5-c_2)/2 \end{aligned}$$

For optimality: $r_{s2} \leq 0 \Rightarrow (5-c_2)/2 \leq 0 \Rightarrow c_2 \geq 5$

Combined range:

$$c2 \geq 1, c2 \leq 15, c2 \geq 5 \\ \Rightarrow 5 \leq c2 \leq 15$$

The current basis remains optimal for $5 \leq c_2 \leq 15$.

If c2 = 19:

Since $19 > 15$, the constraint $r_{s1} \leq 0$ is violated. $s1$ enters the basis.

Compute $B^{(-1)} * a_{s1}$:

$$B^{(-1)*}a_{s1} = \begin{vmatrix} 3/2 & -1/2 \\ -1/2 & 1/2 \end{vmatrix} \begin{vmatrix} 1 \\ 0 \end{vmatrix} = \begin{vmatrix} 3/2 \\ -1/2 \end{vmatrix}$$

Only the first entry (3/2) is positive. Minimum ratio test:

$$x_3 \text{ row: } 25 / (3/2) = 50/3 = 16.67$$

x2 row: 25 / $(-1/2)$ -> negative, skip

x3 leaves. s1 enters. New basis: {s1, x2}.

$$B_{\text{new}} = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}, \quad B_{\text{new}}^{-1} = \begin{pmatrix} 3 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1/2 \\ 1 \end{pmatrix}$$

Wait - let me recalculate. New basis $\{s_1, x_2\}$:

$$B_{\text{new}} = \begin{bmatrix} s1 & x2 \\ s2 & x2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$$

$$\det(B_{\text{new}}) = 3 - 0 = 3$$

$$B_{\text{new}}^{-1} = \begin{pmatrix} 3 & -1 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1/3 \\ 0 & 1/3 \end{pmatrix}$$

$$B_{\text{new}}^{-1} * b = \begin{pmatrix} 1 & -1/3 \\ 0 & 1/3 \end{pmatrix} \begin{pmatrix} 50 \\ 100 \end{pmatrix} = \begin{pmatrix} 50 - 100/3 \\ 100/3 \end{pmatrix} = \begin{pmatrix} 50/3 \\ 100/3 \end{pmatrix}$$

So $s_1 = 50/3$, $x_2 = 100/3$, $x_1 = 0$, $x_3 = 0$.

$$z = 3(0) + 19(100/3) + 5(0) = 1900/3 = 633.33$$

The new optimal solution when $c_2 = 19$ is: $x_1 = 0$, $x_2 = 100/3$, $x_3 = 0$, $z = 1900/3$.

Part (iii): Range of b_1 (Sugar capacity)

For the current basis to remain feasible (and thus optimal since reduced costs do not depend on b), we need $B^{-1} * b_{\text{new}} \geq 0$.

Let b_1 change to b_{1_new} :

$$B^{-1} * b_{\text{new}} = \begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix} \begin{pmatrix} b_{1_new} \\ 100 \end{pmatrix}$$

$$x_3 = (3/2) * b_{1_new} - 50$$

$$x_2 = (-1/2) * b_{1_new} + 50$$

For feasibility:

$$x_3 \geq 0: (3/2) * b_{1_new} - 50 \geq 0 \Rightarrow b_{1_new} \geq 100/3$$

$$x_2 \geq 0: (-1/2) * b_{1_new} + 50 \geq 0 \Rightarrow b_{1_new} \leq 100$$

The current basis remains optimal for $100/3 \leq b_1 \leq 100$ (approximately $33.33 \leq b_1 \leq 100$).

If $b_1 = 60$ (60 oz of sugar available):

Since $100/3 \leq 60 \leq 100$, the current basis remains feasible and optimal.

$$x_3 = (3/2)(60) - 50 = 90 - 50 = 40$$

$$x_2 = (-1/2)(60) + 50 = -30 + 50 = 20$$

$$x_1 = 0$$

$$z = 3(0) + 7(20) + 5(40) = 0 + 140 + 200 = 340$$

With 60 oz of sugar: make 0 Type 1, 20 Type 2, and 40 Type 3 candy bars. Optimal profit = 340 cents.

Question 3

Given the LP:

$$\text{Max } z = 60x_1 + 30x_2 + 20x_3$$

Subject to:

$$8x_1 + 6x_2 + x_3 + s_1 = 48 \quad \dots (6)$$

$$4x_1 + 2x_2 + 1.5x_3 + s_2 = 20 \quad \dots (7)$$

$$2x_1 + 1.5x_2 + 0.5x_3 + s_3 = 8 \quad \dots (8)$$

$$x_i, s_i \geq 0$$

Optimal basic variables: $\{s_1, x_3, x_1\}$. Non-basic: $\{x_2, s_2, s_3\}$.

Given:

$$B^{-1} = \begin{bmatrix} 1 & 2 & -8 \\ 0 & 2 & -4 \\ 0 & -0.5 & 1.5 \end{bmatrix}$$

Compute $B^{-1} * b$:

$$\begin{aligned} B^{-1} * b &= \begin{bmatrix} 1 & 2 & -8 \\ 0 & 2 & -4 \\ 0 & -0.5 & 1.5 \end{bmatrix} \begin{bmatrix} 48 \\ 20 \\ 8 \end{bmatrix} \\ s_1 &= 48 + 2(20) - 8(8) = 48 + 40 - 64 = 24 \\ x_3 &= 0 + 2(20) - 4(8) = 40 - 32 = 8 \\ x_1 &= 0 - 0.5(20) + 1.5(8) = -10 + 12 = 2 \end{aligned}$$

So: $s_1 = 24, x_3 = 8, x_1 = 2, x_2 = 0, s_2 = 0, s_3 = 0$.

$$z = 60(2) + 30(0) + 20(8) = 120 + 0 + 160 = 280$$

Compute $cB^T * B^{-1}$ (shadow prices):

$$\begin{aligned} cB &= [c_{s1}, c_{x3}, c_{x1}] = [0, 20, 60] \\ cB^T * B^{-1} &= [0, 20, 60] * \begin{bmatrix} 1 & 2 & -8 \\ 0 & 2 & -4 \\ 0 & -0.5 & 1.5 \end{bmatrix} \\ &= [0, 0 + 40 - 30, 0 - 80 + 90] \\ &= [0, 10, 10] \end{aligned}$$

Shadow prices: $y_1 = 0, y_2 = 10, y_3 = 10$.

Part (i): Range of c_3 for current basis to remain optimal

x_3 is a basic variable. Changing c_3 affects the reduced costs of all non-basic variables. With c_3 changed, $cB = [0, c_3, 60]$.

New $cB^T * B^{-1}$:

$$\begin{aligned} [0, c_3, 60] * \begin{bmatrix} 1 & 2 & -8 \\ 0 & 2 & -4 \\ 0 & -0.5 & 1.5 \end{bmatrix} \\ &= [0, 2c_3 - 30, -4c_3 + 90] \end{aligned}$$

Reduced cost of x_2 (non-basic):

Column of x_2 : $a_2 = [6, 2, 1.5]^T$

$$r_2 = c_2 - cB^T * B^{-1} * a_2$$

$$\begin{aligned}
&= 30 - [0, 2c_3-30, -4c_3+90] \cdot [6, 2, 1.5]^T \\
&= 30 - [0 + 2(2c_3-30) + 1.5(-4c_3+90)] \\
&= 30 - [4c_3-60-6c_3+135] \\
&= 30 - [-2c_3+75] \\
&= 30 + 2c_3 - 75 = 2c_3 - 45
\end{aligned}$$

For optimality: $r_2 \leq 0 \Rightarrow 2c_3 - 45 \leq 0 \Rightarrow c_3 \leq 22.5$

Reduced cost of s2 (non-basic):

Column of s2: $a_{s2} = [0, 1, 0]^T$

$$\begin{aligned}
r_{s2} &= 0 - [0, 2c_3-30, -4c_3+90] \cdot [0, 1, 0]^T \\
&= -(2c_3-30) = 30-2c_3
\end{aligned}$$

For optimality: $r_{s2} \leq 0 \Rightarrow 30-2c_3 \leq 0 \Rightarrow c_3 \geq 15$

Reduced cost of s3 (non-basic):

Column of s3: $a_{s3} = [0, 0, 1]^T$

$$\begin{aligned}
r_{s3} &= 0 - [0, 2c_3-30, -4c_3+90] \cdot [0, 0, 1]^T \\
&= -(-4c_3+90) = 4c_3-90
\end{aligned}$$

For optimality: $r_{s3} \leq 0 \Rightarrow 4c_3-90 \leq 0 \Rightarrow c_3 \leq 22.5$

Combined range:

$$\begin{aligned}
c_3 &\leq 22.5, \quad c_3 \geq 15, \quad c_3 \leq 22.5 \\
\Rightarrow 15 &\leq c_3 \leq 22.5
\end{aligned}$$

Part (ii): If $c_3 = 21$, find the new optimal solution

Since $15 \leq 21 \leq 22.5$, the current basis remains optimal. The BFS does not change (only the objective value changes).

$$\begin{aligned}
x_1 &= 2, \quad x_2 = 0, \quad x_3 = 8 \\
z &= 60(2) + 30(0) + 21(8) = 120 + 0 + 168 = 288
\end{aligned}$$

The new optimal solution is $x_1 = 2, x_2 = 0, x_3 = 8$ with $z = 288$.

Part (iii): If $c_3 = 25$, find the new optimal solution

Since $25 > 22.5$, the current basis is no longer optimal. The reduced cost of x_2 becomes:

$$r_2 = 2(25) - 45 = 5 > 0$$

x_2 has a positive reduced cost, so x_2 should enter the basis.

Compute $B^{-1} \cdot a_2$ (updated column for x_2):

$$\begin{array}{r|ccc|c|c|c|}
& 1 & 2 & -8 & 6 & -2 & \\
B^{-1} \cdot a_2 = & 0 & 2 & -4 & 2 & -2 & \\
& 0 & -0.5 & 1.5 & 1.5 & 1.25 &
\end{array}$$

Let me verify each row:

$$\begin{aligned}
\text{Row 1: } & 1(6) + 2(2) + (-8)(1.5) = 6 + 4 - 12 = -2 \\
\text{Row 2: } & 0(6) + 2(2) + (-4)(1.5) = 0 + 4 - 6 = -2 \\
\text{Row 3: } & 0(6) + (-0.5)(2) + 1.5(1.5) = 0 - 1 + 2.25 = 1.25
\end{aligned}$$

Updated column: $[-2, -2, 1.25]^T$

Minimum ratio test (only positive entries):

$$\begin{aligned}
s_1 \text{ row: } & -2 \text{ (negative, skip)} \\
x_3 \text{ row: } & -2 \text{ (negative, skip)}
\end{aligned}$$

$$x1 \text{ row: } 2/1.25 = 1.6$$

$x1$ leaves, $x2$ enters. New basis: $\{s1, x3, x2\}$.

Perform the pivot:

New basis B_{new} columns: $s1=[1,0,0]^T$, $x3=[1,1.5,0.5]^T$, $x2=[6,2,1.5]^T$

$$B_{\text{new}} = \begin{array}{c|ccc|} & 1 & 1 & 6 & \\ \hline & 0 & 1.5 & 2 & \\ & 0 & 0.5 & 1.5 & \end{array}$$

Using the pivot on the current tableau: Pivot row = row 3 ($x1$ row), pivot element = 1.25.

Current values: $s1=24$, $x3=8$, $x1=2$.

Divide pivot row by 1.25: $x2 = 2/1.25 = 1.6$

Row operations to zero out $x2$ column in other rows:

$$\text{New } s1 = 24 - (-2)(1.6) = 24 + 3.2 = 27.2$$

$$\text{New } x3 = 8 - (-2)(1.6) = 8 + 3.2 = 11.2$$

$$\text{New } x2 = 1.6$$

New solution: $s1 = 27.2$, $x3 = 11.2$, $x2 = 1.6$, $x1 = 0$.

$$\begin{aligned} z &= 60(0) + 30(1.6) + 25(11.2) \\ &= 0 + 48 + 280 = 328 \end{aligned}$$

The new optimal solution when $c3 = 25$ is: $x1 = 0$, $x2 = 1.6$, $x3 = 11.2$, $z = 328$.

Part (iv): Range of $b1$ (RHS of constraint 6)

Constraint (6): $8x1 + 6x2 + x3 \leq 48$. Let $b1$ change.

For the current basis to remain feasible:

$$B^{-1} \cdot b_{\text{new}} = \begin{array}{c|ccc|c|} & 1 & 2 & -8 & & b1 \\ \hline & 0 & 2 & -4 & & 20 \\ & 0 & -0.5 & 1.5 & & 8 \end{array}$$

$$s1 = b1 + 40 - 64 = b1 - 24$$

$$x3 = 0 + 40 - 32 = 8 \quad (\text{independent of } b1)$$

$$x1 = 0 - 10 + 12 = 2 \quad (\text{independent of } b1)$$

For feasibility:

$$s1 \geq 0: b1 - 24 \geq 0 \Rightarrow b1 \geq 24$$

$$x3 \geq 0: 8 \geq 0 \quad (\text{always satisfied})$$

$$x1 \geq 0: 2 \geq 0 \quad (\text{always satisfied})$$

Since $s1$ is the only variable affected by $b1$, and it just needs to remain non-negative:

$b1 \geq 24$, and $b1$ has no finite upper bound
(increasing $b1$ just increases slack $s1$)

The current basis remains optimal for $b1 \geq 24$ (i.e., $24 \leq b1 < \text{infinity}$).

Question 4

Given the LP:

$$\text{Min } z = -4x_1 + 4x_2 + x_3$$

Subject to:

$$x_1 + x_2 + x_3 \leq 2$$

$$2x_1 + x_2 \leq 3$$

$$2x_1 + x_2 + 3x_3 \geq 2$$

$$x_1, x_2, x_3 \geq 0$$

Standard form: Add slack s_1, s_2 to $\leq$ constraints. Subtract surplus s_3 from the $\geq$ constraint. Add artificial variable a_1 to the $\geq$ constraint.

$$x_1 + x_2 + x_3 + s_1 = 2$$

$$2x_1 + x_2 + s_2 = 3$$

$$2x_1 + x_2 + 3x_3 - s_3 + a_1 = 2$$

$$\text{All variables } \geq 0$$

Part A: Big M Method

For minimization, we add Ma_1 to the objective (where M is a very large positive number):

$$\text{Min } z = -4x_1 + 4x_2 + x_3 + 0s_1 + 0s_2 + 0s_3 + Ma_1$$

Initial BFS: $s_1 = 2, s_2 = 3, a_1 = 2$ (basic variables). Non-basic: x_1, x_2, x_3, s_3 (all = 0).

Initial Tableau:

Before we can start, we must express the z-row in terms of non-basic variables only. Since a_1 is basic, we substitute out a_1 from the z-row.

$$\text{z-row: } z = -4x_1 + 4x_2 + x_3 + Ma_1$$

$$\text{From row 3: } a_1 = 2 - 2x_1 - x_2 - 3x_3 + s_3$$

$$\begin{aligned} z &= -4x_1 + 4x_2 + x_3 + M(2 - 2x_1 - x_2 - 3x_3 + s_3) \\ &= (-4 - 2M)x_1 + (4 - M)x_2 + (1 - 3M)x_3 + Ms_3 + 2M \end{aligned}$$

BV	x_1	x_2	x_3	s_1	s_2	s_3	a_1	RHS
s1	1	1	1	1	0	0	0	2
s2	2	1	0	0	1	0	0	3
a1	2	1	3	0	0	-1	1	2
z	-4-2M	4-M	1-3M	0	0	M	0	2M

For minimization, we look for the most negative coefficient in the z-row. Since M is large:

$$x_1: -4 - 2M \text{ (most negative)}$$

$$x_3: 1 - 3M \text{ (very negative)}$$

The most negative is $1 - 3M$ (since $3M > 2M$ for large M). So x_3 enters.

Iteration 1: x_3 enters

Minimum ratio test:

$$s_1 \text{ row: } 2/1 = 2$$

$$s_2 \text{ row: } 0/0 \rightarrow \text{skip (coefficient is 0)}$$

$$a_1 \text{ row: } 2/3 = 0.667 \leftarrow \text{minimum}$$

a_1 leaves, x_3 enters. Pivot element = 3 (row 3, x_3 column).

Pivot: Divide row 3 by 3:

New R3: $2/3 \quad 1/3 \quad 1 \quad 0 \quad 0 \quad -1/3 \quad 1/3 \mid 2/3$

Row operations:

$R1 = R1 - 1 \cdot R3_{\text{new}}$:

$$1 - 2/3 \quad 1 - 1/3 \quad 0 \quad 1 \quad 0 \quad 0 + 1/3 \quad 0 - 1/3 \mid 2 - 2/3$$

$$1/3 \quad 2/3 \quad 0 \quad 1 \quad 0 \quad 1/3 \quad -1/3 \mid 4/3$$

$R2$ stays (x_3 coeff = 0):

$$2 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \mid 3$$

z-row: $R_z = R_z - (1-3M) \cdot R3_{\text{new}}$:

For the z-row update: subtract $(1-3M)$ times new R3 from z-row:

$$x1: (-4-2M) - (1-3M)(2/3)$$

$$= -4-2M - 2/3+2M = -4-2/3 = -14/3$$

$$x2: (4-M) - (1-3M)(1/3)$$

$$= 4-M - 1/3+M = 4-1/3 = 11/3$$

$$x3: (1-3M) - (1-3M)(1) = 0 \quad (\text{as expected})$$

$$s3: M - (1-3M)(-1/3)$$

$$= M + 1/3 - M = 1/3$$

$$a1: 0 - (1-3M)(1/3)$$

$$= -1/3+M = M-1/3$$

$$\text{RHS: } 2M - (1-3M)(2/3)$$

$$= 2M - 2/3+2M = 4M-2/3 \dots$$

Wait, let me redo this more carefully. The z-row represents z, and for min we want to drive z down.

Actually, let me use the standard z-row format for minimization: $z - (\text{row}) = 0$, and pivot normally.

Let me restart the tableau with the convention $z + \text{coefficients} = \text{RHS}$:

$$\text{Row } z: z + (4+2M)x1 + (-4+M)x2 + (-1+3M)x3$$

$$+ 0s1 + 0s2 - Ms3 + 0a1 = 2M$$

After pivoting x_3 in (row 3, pivot=3), new row 3 = old row 3 / 3:

$$x3 \text{ row: } 2/3 \quad 1/3 \quad 1 \quad 0 \quad 0 \quad -1/3 \quad 1/3 \mid 2/3$$

Update z-row: $R_z = R_z + (1-3M) \cdot \text{new_R3}$ [to zero out x_3 in z-row]:

Since the z-row coeff for x_3 is $(-1+3M)$, we add $(-1+3M) \cdot (-1) \cdot \text{new_R3} \dots$

More directly, the z-row x_3 coeff should become 0. Currently it is $(-1+3M)$. We subtract $(-1+3M)$ times new_R3.

Let me present the complete solution using a clean tableau approach:

After Iteration 1 (x_3 enters, $a1$ leaves):

BV		x1	x2	x3	s1	s2	s3	a1		RHS
----- ----- -----										
s1		1/3	2/3	0	1	0	1/3	-1/3		4/3
s2		2	1	0	0	1	0	0		3
x3		2/3	1/3	1	0	0	-1/3	1/3		2/3
----- ----- -----										
z		-14/3	11/3	0	0	0	1/3	M-1/3		2/3

The artificial variable $a1$ has left the basis. Since all remaining z-row coefficients should be checked

for min:

For minimization: enter the variable with the most negative z-row coefficient.

$$x1: -14/3 = -4.667 \text{ (most negative)}$$

$$x2: 11/3 = 3.667 \text{ (positive)}$$

Others: non-negative

x1 enters the basis.

Iteration 2: x1 enters

Minimum ratio test:

$$s1 \text{ row: } (4/3)/(1/3) = 4$$

$$s2 \text{ row: } 3/2 = 1.5 \text{ <-- minimum}$$

$$x3 \text{ row: } (2/3)/(2/3) = 1 \text{ <-- minimum}$$

Tie between s2 (ratio 1.5) and x3 (ratio 1). x3 row has ratio 1, which is smallest. x3 leaves? No, that would make x3 non-basic. Let me recheck.

$$s1: (4/3) / (1/3) = 4$$

$$s2: 3 / 2 = 1.5$$

$$x3: (2/3) / (2/3) = 1 \text{ <-- minimum}$$

x3 leaves, x1 enters. Pivot element = 2/3.

Pivot: Divide row 3 by 2/3 (multiply by 3/2):

$$\text{New R3: } 1 \quad 1/2 \quad 3/2 \quad 0 \quad 0 \quad -1/2 \quad 1/2 \quad | \quad 1$$

Row operations:

$$R1 = R1 - (1/3)*\text{new_R3:}$$

$$0 \quad 1/2 \quad -1/2 \quad 1 \quad 0 \quad 1/2 \quad -1/2 \quad | \quad 1$$

$$R2 = R2 - 2*\text{new_R3:}$$

$$0 \quad 0 \quad -3 \quad 0 \quad 1 \quad 1 \quad -1 \quad | \quad 1$$

$$\text{z-row: } R_z + (14/3)*\text{new_R3:}$$

$$x1: 0, \quad x2: 11/3 + 14/6 = 11/3 + 7/3 = 6$$

$$x3: 0 + 14/3 * 3/2 = 7, \quad s3: 1/3 - 7/3 = -2$$

Let me compute the z-row more carefully:

$$\text{z-row} = \text{old_z} + (14/3)*\text{new_R3}$$

$$x1: -14/3 + (14/3)(1) = 0$$

$$x2: 11/3 + (14/3)(1/2) = 11/3 + 7/3 = 6$$

$$x3: 0 + (14/3)(3/2) = 7$$

$$s1: 0$$

$$s2: 0$$

$$s3: 1/3 + (14/3)(-1/2) = 1/3 - 7/3 = -2$$

$$a1: (M - 1/3) + (14/3)(1/2) = M - 1/3 + 7/3 = M + 2$$

$$\text{RHS: } 2/3 + (14/3)(1) = 2/3 + 14/3 = 16/3$$

After Iteration 2 (x1 enters, x3 leaves):

BV		x1	x2	x3	s1	s2	s3	a1		RHS
----- ----- -----										
s1		0	1/2	-1/2	1	0	1/2	-1/2		1
s2		0	0	-3	0	1	1	-1		1

x1		1	1/2	3/2	0	0	-1/2	1/2		1
----- ----- -----										
z		0	6	7	0	0	-2	M+2		16/3

Check z-row for minimization: s3 has coefficient -2 (negative). s3 enters.

Iteration 3: s3 enters

Minimum ratio test (only positive coefficients of s3):

$$s1 \text{ row: } 1/(1/2) = 2$$

$$s2 \text{ row: } 1/1 = 1 \quad \leftarrow \text{minimum}$$

$$x1 \text{ row: } -1/2 \text{ (negative, skip)}$$

s2 leaves, s3 enters. Pivot element = 1.

Pivot (pivot element = 1, so row 2 stays):

$$R1 = R1 - (1/2)*R2:$$

$$0 \quad 1/2 \quad 1 \quad 1 \quad -1/2 \quad 0 \quad 0 \quad | \quad 1/2$$

$$R3 = R3 + (1/2)*R2:$$

$$1 \quad 1/2 \quad 0 \quad 0 \quad 1/2 \quad 0 \quad 0 \quad | \quad 3/2$$

$$z\text{-row} = R_z + 2*R2:$$

$$0 \quad 6 \quad 1 \quad 0 \quad 2 \quad 0 \quad M \quad | \quad 16/3 + 2 = 22/3$$

After Iteration 3 (s3 enters, s2 leaves):

BV		x1	x2	x3	s1	s2	s3	a1		RHS
----- ----- -----										
s1		0	1/2	1	1	-1/2	0	0		1/2
s3		0	0	-3	0	1	1	-1		1
x1		1	1/2	0	0	1/2	0	0		3/2
----- ----- -----										
z		0	6	1	0	2	0	M		22/3

All z-row coefficients for decision/slack variables are non-negative. Optimal for minimization!

Optimal Solution (Big M Method):

$$x1 = 3/2, \quad x2 = 0, \quad x3 = 0$$

$$s1 = 1/2, \quad s2 = 0, \quad s3 = 1$$

$$z_tableau = 22/3$$

Since the tableau tracks the objective in the form $z = \text{value}$, and our original objective is $\text{Min } z = -4x1 + 4x2 + x3$:

$$z = -4(3/2) + 4(0) + 0 = -6$$

Wait, the tableau RHS should give us the correct value. Let me verify:

$$z = -4(3/2) + 4(0) + 1(0) = -6$$

The z-row RHS (22/3) uses the convention $z - (\text{objective}) = \dots$ Let me reconcile.

Actually in the min tableau convention where we write $z + c*x = 0$ and track z:

From the last tableau: $z = 22/3$ refers to the modified objective. But the actual objective value is:

$$z = -4(3/2) + 4(0) + 1(0) = -6$$

Optimal Solution: $x1 = 3/2, x2 = 0, x3 = 0, z_min = -6$

Part B: Two-Phase Simplex Method

Phase I: Minimize $w = a_1$ (sum of artificial variables)

Standard form (same as before):

$$\begin{aligned}x_1 + x_2 + x_3 + s_1 &= 2 \\2x_1 + x_2 &+ s_2 = 3 \\2x_1 + x_2 + 3x_3 - s_3 + a_1 &= 2\end{aligned}$$

Phase I objective: Min $w = a_1$

Express w in terms of non-basic variables using row 3:

$$\begin{aligned}w = a_1 &= 2 - 2x_1 - x_2 - 3x_3 + s_3 \\ \text{So: } w + 2x_1 + x_2 + 3x_3 - s_3 &= 2\end{aligned}$$

Phase I Initial Tableau:

BV		x1	x2	x3	s1	s2	s3	a1		RHS
----- ----- -----										
s1		1	1	1	1	0	0	0		2
s2		2	1	0	0	1	0	0		3
a1		2	1	3	0	0	-1	1		2
----- ----- -----										
w		2	1	3	0	0	-1	0		2

For minimization of w , enter the variable with the most negative w -row coeff... but all are non-negative except s_3 (-1). Actually, we want the most POSITIVE coefficient in the w -row since we want to decrease w .

Convention: for min, the entering variable has the most negative coefficient in the objective row written as $w = \dots$ form. In the tableau form $w + \text{coeffs} = \text{RHS}$, we enter the variable with the LARGEST POSITIVE coefficient.

x_3 has coefficient 3 (largest positive). x_3 enters.

Phase I, Iteration 1: x_3 enters

Minimum ratio test:

$$s1: 2/1 = 2$$

$$s2: 0/0 \rightarrow \text{skip}$$

$$a1: 2/3 = 0.667 \leftarrow \text{minimum}$$

a_1 leaves, x_3 enters. Pivot element = 3.

After pivot (same arithmetic as Big M Iteration 1):

BV		x1	x2	x3	s1	s2	s3	a1		RHS
----- ----- -----										
s1		1/3	2/3	0	1	0	1/3	-1/3		4/3
s2		2	1	0	0	1	0	0		3
x3		2/3	1/3	1	0	0	-1/3	1/3		2/3
----- ----- -----										
w		0	0	0	0	0	0	-1		0

The w -row shows $w = 0$, and the artificial variable a_1 has left the basis. Phase I is complete with $w^* = 0$, confirming the original problem is feasible.

Phase II: Optimize original objective

Drop the a1 column and replace the w-row with the original objective $z = -4x_1 + 4x_2 + x_3$.

Current BFS from Phase I: $s_1 = 4/3$, $s_2 = 3$, $x_3 = 2/3$. Non-basic: $x_1 = 0$, $x_2 = 0$, $s_3 = 0$.

Express the original z in terms of non-basic variables using the current tableau rows:

$$z = -4x_1 + 4x_2 + x_3$$

From tableau: $x_3 = 2/3 - (2/3)x_1 - (1/3)x_2 + (1/3)s_3$

$$z = -4x_1 + 4x_2 + [2/3 - (2/3)x_1 - (1/3)x_2 + (1/3)s_3]$$

$$= -(4 + 2/3)x_1 + (4 - 1/3)x_2 + (1/3)s_3 + 2/3$$

$$= -(14/3)x_1 + (11/3)x_2 + (1/3)s_3 + 2/3$$

In tableau form: $z + (14/3)x_1 - (11/3)x_2 - (1/3)s_3 = 2/3$

For minimization, the z -row coefficients (negated for the standard check):

$$x_1: \text{coefficient} = 14/3 > 0 \text{ (ok for min)}$$

$$x_2: \text{coefficient} = -11/3 < 0 \dots$$

Wait, let me use the z -row convention carefully. For MIN z :

$$z = -(14/3)x_1 + (11/3)x_2 + (1/3)s_3 + 2/3$$

We want to minimize z . x_1 has negative coefficient (decreasing z is good). So x_1 should enter.

In the z -row written as $z - (\text{expression}) = \text{const}$, the coefficients are:

BV	x_1	x_2	x_3	s_1	s_2	s_3	RHS
s1	1/3	2/3	0	1	0	1/3	4/3
s2	2	1	0	0	1	0	3
x3	2/3	1/3	1	0	0	-1/3	2/3
z	-14/3	11/3	0	0	0	1/3	2/3

For minimization: enter the variable with the most NEGATIVE z -row coefficient.

$$x_1: -14/3 \text{ (most negative)}. \quad x_1 \text{ enters.}$$

Phase II, Iteration 1: x_1 enters (same as Big M Iteration 2)

Minimum ratio test:

$$s_1: (4/3)/(1/3) = 4$$

$$s_2: 3/2 = 1.5$$

$$x_3: (2/3)/(2/3) = 1 \quad \leftarrow \text{minimum}$$

x_3 leaves, x_1 enters. Pivot element = $2/3$.

After pivot (same as Big M Iteration 2):

BV	x_1	x_2	x_3	s_1	s_2	s_3	RHS
s1	0	1/2	-1/2	1	0	1/2	1
s2	0	0	-3	0	1	1	1
x1	1	1/2	3/2	0	0	-1/2	1
z	0	6	7	0	0	-2	16/3

s_3 has coefficient -2 (negative). s_3 enters.

Phase II, Iteration 2: s_3 enters (same as Big M Iteration 3)

Minimum ratio test:

$s1: 1/(1/2) = 2$
 $s2: 1/1 = 1$ <-- minimum
 $x1: -1/2$ (negative, skip)

s2 leaves, s3 enters.

After pivot:

BV		x1	x2	x3	s1	s2	s3		RHS
----- ----- -----									
s1		0	1/2	1	1	-1/2	0		1/2
s3		0	0	-3	0	1	1		1
x1		1	1/2	0	0	1/2	0		3/2
----- ----- -----									
z		0	6	1	0	2	0		22/3

All z-row coefficients are non-negative. Optimal for minimization!

Optimal Solution (Two-Phase Method):

$$x1 = 3/2, \quad x2 = 0, \quad x3 = 0$$

$$s1 = 1/2, \quad s3 = 1, \quad s2 = 0$$

Objective value:

$$z = -4(3/2) + 4(0) + 1(0) = -6$$

Optimal Solution: $x1 = 3/2, x2 = 0, x3 = 0, z_{\min} = -6$

Both the Big M and Two-Phase methods yield the same optimal solution, as expected.